

Electric Field due to a point charge:

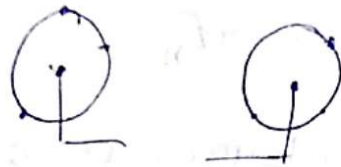
(2)

$$F = \frac{Kq_1q_2}{r^2}$$

⇒ Coulomb said Represent charges as point charges

⇒ If r which is distance is very very large as compared to size of charges then we call them point charges

$$r \gg \text{Size}$$



⇒ Force is a vector then it has a unit vector

$$\vec{F} = K \frac{q_1q_2}{r^2} \hat{r}$$

$$\therefore F = E/q_0$$

$$\therefore E = \frac{\vec{F}}{q_0} \quad \therefore q_0 = \text{Test charge}$$

$$\vec{E} = \frac{Kq_1q_2}{r^2} \hat{r} \div q_0 = \frac{Kq_1}{r^2} \hat{r}$$

$$\boxed{\vec{E} = K \frac{q}{r^2} \hat{r}} \rightarrow \text{electric field due to a point charge}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$K = \frac{1}{4\pi\epsilon_0} \Rightarrow K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

$$E_{\text{med}} = \frac{K}{\epsilon_0} \frac{q}{r^2}$$

⇒ Superposition principle is fundamental in understanding electric fields & force in context of multiple charges

Superposition Principle for Electric Fields:

⇒ The superposition principle states that the net E.F at a point due to several charges is the vector sum of the electric fields created by each individual charge at that point. ~~Mean~~

⇒ Means if there are multiple charges the net electric field E_{net} at a given point is found by adding the electric field vectors $E_1, E_2, \dots, E_n$ due to each charge.

$$E_{net} = E_1 + E_2 + \dots + E_n$$

⇒ Electric field 'E' created by charge q_1 at a distance r from it (for a test charge q_2) is related to the force by the eq $E = \frac{F}{q_2}$

⇒ Vector addition: You need to break the electric fields into components (usually along x & y axes) and add them component wise.

$$E_{net,x} = E_{1x} + E_{2x} + \dots$$

$$E_{net,y} = E_{1y} + E_{2y} + \dots$$

⇒ Magnitude: After adding the components, we can find the total magnitude of net E.F from its components.

$$E_{net} = \sqrt{E_{net,x}^2 + E_{net,y}^2}$$

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Electric field due to an electric dipole:

Consider electric field created by an electric dipole which consists of two opposite charges of equal magnitude, separated by distance d . The

$\Rightarrow$ Field analysed at point 'P' along dipole axis located at a distance z from the dipole's midpoint.

$\Rightarrow$ E.F at point 'P' due to +ve charge $+q$ (located closer) points in the +ve direction along z axis, with magnitude $E(+)$

$\Rightarrow$ E.F at point 'P' due to -ve charge $-q$ (located farther) points in the -ve direction along z -axis with magnitude $E(-)$

$\Rightarrow$ Net E.F at point P is sum of individual fields

$$E_{\text{net}} = E(+) + E(-)$$

$\Rightarrow$ The fields are along the same axis, their direction (signs)

are critical when summing them. The net field is found by combining the contributions from both charges.

$\Rightarrow$ The product of charge q & the separation distance d is called dipole moment P

$$E = E_+ - E_-$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r_+^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_-^2}$$

$$= \frac{q}{4\pi\epsilon_0 (z - \frac{d}{2})^2} - \frac{q}{4\pi\epsilon_0 (z + \frac{d}{2})^2}$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{1}{(1 - \frac{d}{2z})^2} - \frac{1}{(1 + \frac{d}{2z})^2} \right)$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{1 - (\frac{d}{2z})^2}{(1 - (\frac{d}{2z})^2)^2} - \frac{1 - (\frac{d}{2z})^2}{(1 + (\frac{d}{2z})^2)^2} \right)$$

$\therefore$ neglect $\frac{d}{2z}$

$$E = \frac{1}{4\pi\epsilon_0} \frac{qd}{z^3}$$

⇒ At large distances we have $d/z \ll 1$. Thus in our approximation we can neglect the $\frac{d}{z}$ in denominator.

Eric

⇒ Product q/d which involves the two intrinsic properties q & d of the dipole is the magnitude P of a vector quantity known as electric dipole moment $\vec{P}$ of the dipole.
Unit of P Coulomb-meter

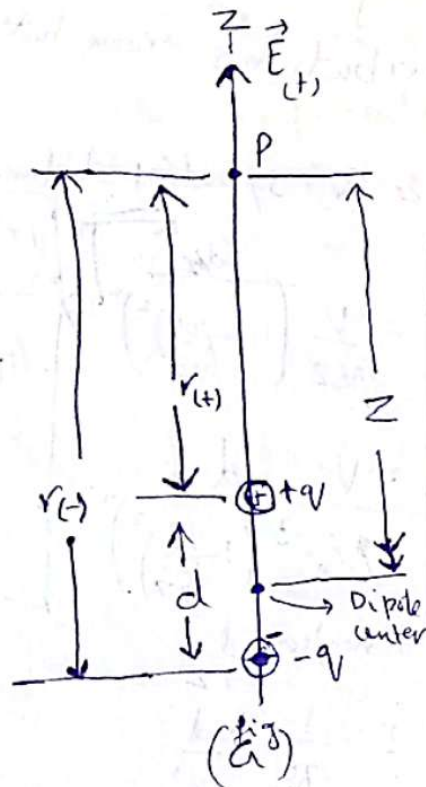
So

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{z^3}$$

Ch

A'

Physical Interpretation: At large distances, a dipole's electric field decreases rapidly because the opposing charges nearly cancel each other out. The residual field falls off as $\frac{1}{r^3}$, reflecting the dipole's separation, unlike the $\frac{1}{r^2}$ decay of a point charge.



up here +q field dominates
(+)

$\vec{P}$

(-)
Down here the -q field dominates

(Fig b)

The dipole moment $\vec{P}$ of the dipole points from -ve charge to the +ve charge

Electric Field Due to a line of charge :

Ring of charges

Charge distribution:

$$\text{As } E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Here $dq = \lambda ds \Rightarrow$ [surface charge density distribution]
 'ds' be arc of length of that element then in terms of linear density λ .
 λ (The charge per unit length).

$$\text{So } dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

x by $\cos\theta$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \cdot \cos\theta \quad \text{--- (A)}$$

Now from the figure, show that

$$\cos\theta = \frac{\text{Base}}{\text{Hyp}} = \frac{z}{r}$$

By Pathagoras theorem (hypotenuse² = Base² + height²)

$$r^2 = z^2 + R^2 \quad r = \sqrt{z^2 + R^2}$$

Putting values in (A)

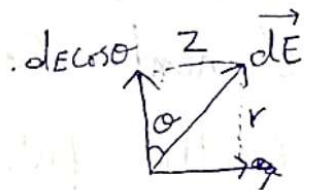
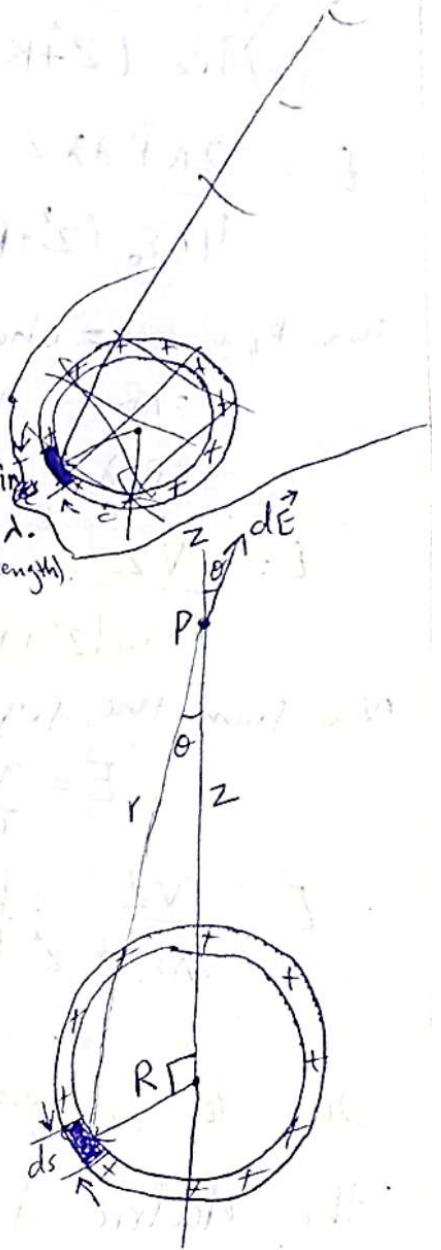
$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{(z^2 + R^2)} \cdot \frac{z}{R}$$

Taking integral from both side, So during this process $S = 0 \rightarrow 2\pi R$, Put value of dq also

$$\int dE = \int_0^{2\pi R} \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(z^2 + R^2)} \cdot \frac{z}{R}$$

$$\int dE = \frac{1}{4\pi\epsilon_0} \frac{1}{(z^2 + R^2)} \cdot \frac{z}{R} \int ds$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{1}{(z^2 + R^2)} \frac{z}{R} \left[S \right]_0^{2\pi R} \Rightarrow \text{Put value of } r \text{ \& opening integral}$$



Hypotenuse is the side opposite the Right angle.

=> Base & height are the sides from the right angle

=> The components perpendicular to z axis cancel, the parallel components add.

=> integral that moves along the ring from element to element from start point $S=0$ through full circumference $S=2\pi R$.

$$= E = \frac{1}{4\pi\epsilon_0} \frac{\lambda z}{(z^2 + R^2)^{3/2}} \left[2\pi R - 0 \right] \Rightarrow \therefore \frac{3}{2} \text{ Came from Square, and other term has power 2}$$

$$E = \frac{2\pi R \lambda z}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

$$(z^2 + R^2)^{3/2} = \sqrt{z^2 + R^2}^3$$

$$(z^2 + R^2)^{-1} (z^2 + R^2)^{1/2} = (z^2 + R^2)^{-1/2} = \frac{1}{\sqrt{z^2 + R^2}}$$

We know λ = charge of unit length

$2\pi R$ = Area of the Ring

$2\pi R \lambda$ = Total charge of the Ring

So $2\pi R \lambda = q$

$$\therefore \lambda = \frac{q}{2\pi R}$$

$$E = \frac{qz}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \rightarrow \text{charged ring.}$$

Now from the figure $z^2 \gg R^2$ so R^2 is neglected

$$E = \frac{qz}{4\pi\epsilon_0 (z^2)^{3/2}}$$

$$E = \frac{qz}{4\pi\epsilon_0 z^3} = \boxed{E = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2}} \rightarrow \text{charged ring at large distance}$$

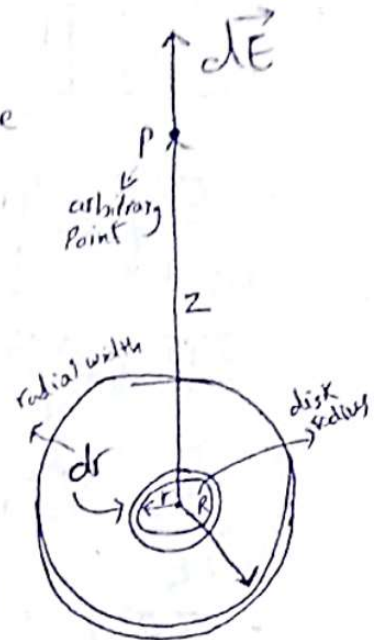
Due to the symmetry of the uniformly charged ring, the electric field contributions from different charge elements cancel out at the center. As a result, the net electric field at the center of the ring is zero. Consequently, no force acts on a test charge placed there. This is why field at the center of a uniformly charged ring is zero, a result that can be generalized to similar systems with high symmetry.

Electric field due to charged disk:

(5)

→ Switch from line of charge to a surface of charge by examining E-F

→ thin ring of radius r & thickness dr on disk carries a small amount of charge dq . The surface charge density is σ , so charge on ring is
 $dq = \sigma dA$
 $dq = \sigma \cdot (2\pi r dr)$



→ $2\pi r dr$ is circumference of ring. → dr radial width of ring
 → dA is differential area of ring
 → Electric field at Point P on central axis (z-axis)

Surface charge density is the amount of electric charge per unit area of a surface. area is (length x width)
 → Arbitrary point: random point that is not specified.

$$\Rightarrow E = \frac{qz}{4\pi\epsilon_0(z^2 + r^2)^{3/2}} \Rightarrow dE = \frac{dqz}{4\pi\epsilon_0(z^2 + r^2)^{3/2}}$$

arbitrary axis $r \leq R$. The ring is so thin that we can treat the charge on it as charge element dq . To find its small contribution dE to the electric field at point P. This in terms of ring's charge dq & radius r .
 ⇒ The ring field points in the +ve direction of the z-axis or

Field is directed along the z axis

$$dE = \frac{Z dq}{4\pi\epsilon_0(z^2 + r^2)^{3/2}}$$

$$dE = \frac{Z(\sigma \cdot 2\pi r dr)}{4\pi\epsilon_0(z^2 + r^2)^{3/2}}$$

$$dE = \frac{Z \sigma r dr}{2\epsilon_0(z^2 + r^2)^{3/2}}$$

$$\int dE = \left[\frac{Z\sigma}{2\epsilon_0} \right] \int_0^R \frac{r}{(z^2 + r^2)^{3/2}} dr$$

values constant that's why they don't exist in integral

$$E = \frac{Z\sigma}{2\epsilon_0} \int_0^R \frac{r}{(z^2 + r^2)^{3/2}} dr$$

x & ÷ by 2 b/c derivative of this is 2r we have r but not 2. So balance the integral

$$E = \frac{Z\sigma}{2\epsilon_0} \frac{1}{2} \int_0^R \frac{2r}{(z^2 + r^2)^{3/2}} dr$$

Integration b/w variable
 r/r=0 & r=R & z remains constant, during this process

$$E = \frac{\sigma}{2\epsilon_0} \frac{1}{2} \left[\frac{(z^2 + r^2)^{-\frac{3}{2} + 1}}{-\frac{3}{2} + 1} \right]_0^R$$

$$= \frac{\sigma}{4\epsilon_0} \left[\frac{(z^2 + r^2)^{-\frac{1}{2}}}{-\frac{1}{2}} \right]_0^R$$

$$= \frac{\sigma}{4\epsilon_0} (-2) \left[(z^2 + r^2)^{-\frac{1}{2}} \right]_0^R$$

$$= -\frac{\sigma}{2\epsilon_0} \left[(z^2 + R^2)^{-\frac{1}{2}} - (z^2 + 0)^{-\frac{1}{2}} \right]$$

$$= -\frac{\sigma}{2\epsilon_0} \left[\frac{1}{\sqrt{z^2 + R^2}} - \frac{1}{\sqrt{z^2}} \right]$$

$$= -\frac{\sigma}{2\epsilon_0} \left[\frac{1}{\sqrt{z^2 + R^2}} - \frac{1}{z} \right]$$

$$= -\frac{\sigma}{2\epsilon_0} \left[\frac{z}{z^2 + R^2} - \frac{z}{z} \right]$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + R^2}} \right]} \rightarrow \text{Charged disk}$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}} \rightarrow \text{infinite sheet}$$

$\Rightarrow$ If we let $z \rightarrow 0$, keeping R finite

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A point charge in an electric field:

- ⇒ Force on charged particle in E.F given by $\vec{F} = q\vec{E}$ & direction of force depends on the sign of charge.
- ⇒ Millikan's oil drop experiment demonstrated the quantization of charge & measured the elementary/single charge e
- ⇒ Ink jet printing uses electric fields to control the placement of charged ink drops on paper.
- ⇒ Electrical breakdown occurs when the electric field in air exceeds a critical value, leading to sparking as electrons are accelerated & collide with air molecules.
- ⇒ As ^{or} electrons are accelerated by the E.F, they collide with air molecules, causing the emission of light. This visible path of ionized air is called a spark.
- ⇒ Direction of electrostatic force on the e^- is opposite to the direction of the electric field.
- ⇒ The e^- will accelerate in the direction opposite to its initial velocity if it was moving parallel to axis
- ⇒ If the e^- is initially moving rightward, its speed in the rightward direction will remain constant.

A Dipole in E-F.

⇒ Electric dipole moment: is a vector quantity that points from the -ve charge to the +ve charge of dipole. A dipole consisting two charges $+q$ & $-q$ separated by a distance d .

$$p = qd$$

where d is the displacement vector from -ve to +ve charge.

⇒ Torque on dipole in E-F:

$$F_{\text{net}} = F_+ + F_- \\ = qE - qE = 0$$

when electric dipole is placed in uniform external field E , each charge experience force. The force on -ve & +ve charge ($F_+ = qE$, $F_- = -qE$) since the field is uniform so net force on dipole is zero.

However force produce Torque τ that tends to rotate the dipole. Torque given as

$$\tau = p \times E$$

magnitude of τ is

$$\tau = pE \sin \theta$$

θ is the angle b/w dipole moment ' p ' & E-F ' E '

Potential Energy of Dipole in E.F. orientation relative to field

$$\begin{aligned} \text{potential energy } U &= -p \cdot E \\ U &= -pE \cos \theta \end{aligned}$$

when $\theta = 0$

$$U = -pE$$

dipole is aligned ^{with the} field, minimum potential energy

• when $\theta = 90^\circ$

$$U = 0$$

• when $\theta = 180^\circ$ Dipole is antialigned with field, maximum P.E

$$U = pE$$

Work done on Dipole: when dipole rotates to initial orientation θ_i to final orientation θ_f , the work done by E.F is equal to -ve changes in P.E

$$W = -\Delta U = -(U_f - U_i)$$

If external agent (like an applied torque) causes the dipole to rotate, the work done by the external agent

$$W_a = \Delta U = U_f - U_i$$